

Proposed March 2026

Project Title: Statement of Work (SOW) for puncture, tear, and tensile testing of fabrics and polymers at cryogenic temperatures

Background: The overarching goal of the NASA's Mars GTU 2020 Project, conducted by the Crew and Thermal System Division Advanced Suit Team, is to characterize current spacesuit materials under a simulated Mars environment. This characterization is twofold, focusing on mechanical and spectroscopy. This statement of work is focused on mechanical only.

For this reason, the Mars GTU 2020 project team is seeking the capability to characterize puncture, tear, and tensile properties of fabrics and tensile properties of polymers all at Mars like cryogenic temperature conditions. This will create the baseline of current suit materials for future down selection.

Objective: Vendor will receive various test specimens, pre-cut to size, per specified ASTM standards. Then, measurements are to be made using the United Testing System (UTS) STN-100KN system at NASA specified temperature conditions.

Period of Performance:

The Period of Performance (POP) for this effort shall be up to 12 months following Approved to Proceed (ATP). The work will be performed on an "as needed" basis when the NASA project team has specimens ready for testing.

Scope of Work:

This scope of work shall specifically address requirements related to fabric and durable plastic for physical degradation and material durability. All work shall be performed in close coordination with NASA to ensure alignment with project objectives.

Deliverables

1. **Check in samples:** Inspect for damage, photograph, label and store in a secure environment.
2. **Specimen preparation:** Verify dimensions of specimens in 'as received' condition. Provide in table format
 - a. Specimen ID – individual tested item
 - b. Sample ID – specific group of specimens
 - c. Sample Description

- d. Length (mm)
 - e. Width (mm)
 - f. Thickness (mm)
3. **Specimen Conditioning:** Monitor temperature and humidity prior to test with thermocouple. Allow specimen to reach a stable temperature and maintain for 30 minutes prior to test. Remove thermocouple before test.
4. **Test Procedure:** Test type and target temperature will be specified by NASA prior to specimen shipment.

1. Test Types:

- a. Total specimen count: 166
- b. ASTM D5035-14, “Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method)”
 - 1. Quantity: 80
- c. ASTM D2261-13, “Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure (Constant-Rate-of-Extension Tensile Testing Machine)”
 - 1. Quantity: 24
- d. ASTM F1342/F1342M-05, “Standard Test Method for Protective Clothing Material Resistance to Puncture”
 - 1. Quantity: 32
- e. ASTM D368-22, “Standard Test Method for Tensile Properties of Plastics”
 - 1. Quantity: 30

2. Target Test Conditions and Output

- a. Temperatures:
 - 1. Conditioning ambient: 23 °C / 73 °F
 - 2. Test Temperature: Cryo: -143 °C / -225 °F
- b. Log force with load cell and log temperature with thermocouple input module.

- c. Compute average and maximum force required for each specimen type tested.
 - d. Compute average and maximum elongation required for each specimen type tested.
 - e. Provide compiled stress versus strain curves per sample ID.
 - f. Collect at minimum 100 data points
 - g. Provide photographs and videos of test setup and test results deemed of interest.
 - h. Provide final report with table of test results and photos.
 - i. Return all tested material to NASA.
- 3. Final report documentation:** with table of test results and photos for each specimen and each test type.
- 4. Return of tested samples:** Shipping requirements as follows for post-test samples to the following address:

NASA LYNDON B. JOHNSON SPACE CENTER
BLDG.420, CENTRAL RECEIVING
2101 NASA PARKWAY
HOUSTON, TEXAS 77058
Delivery to: Bldg.7A/RM-322
Attn: Kristine Davis, Project Code-02226

Travel:

No travel will be required for this work.

Contract Type:

The services described in this Statement of Work shall be provided on a Firm Fixed Price (FFP) basis. Vendor shall perform the services using commercially reasonable best efforts and in accordance with the terms and conditions of the governing Agreement. The fixed price is not subject to adjustment based on the Contractor's actual costs incurred in performing the services.